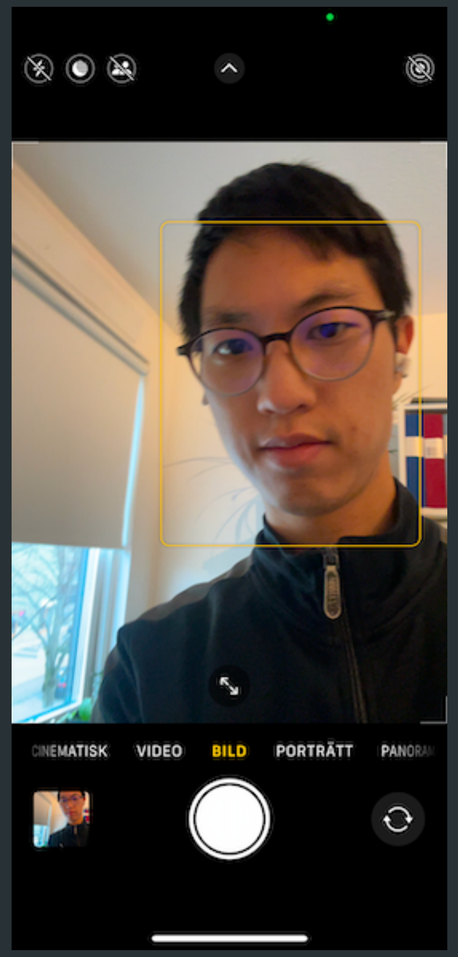
AI-genererad bild med flux. 1 schnell

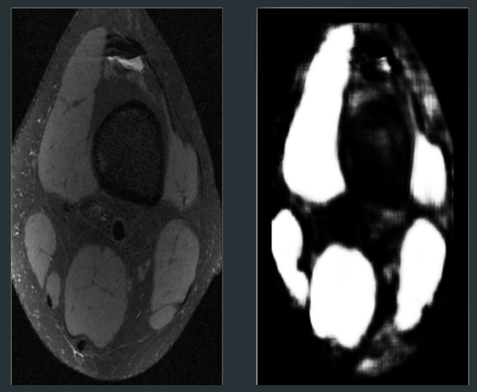
kokchun giang

a gentle introduction to what **AI** is, its history and how to approach an AI project

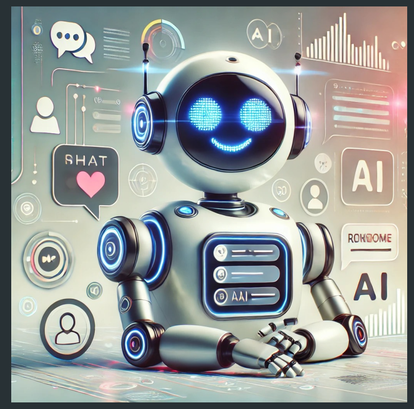
a few examples of **AI applications**



facial detection using phone camera

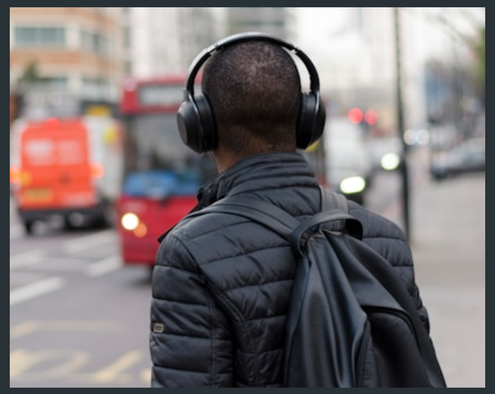


3D segmentation of MRI



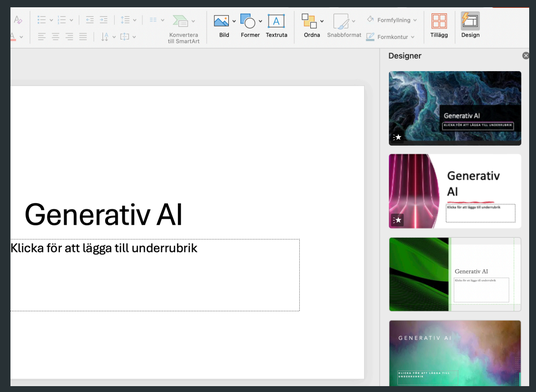
chatbots (AI-genererad with Dall-E 3)

spam filter in email



recommender system for music

targeted digital advertisement



design suggestions in powerpoint

AI-genererad bild med flux. 1 schnell

a few examples of **generative AI** for images and text

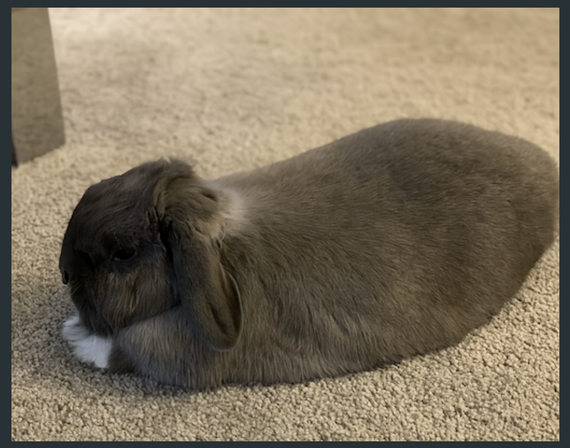
where is **Bella?**

a)

b)

c)

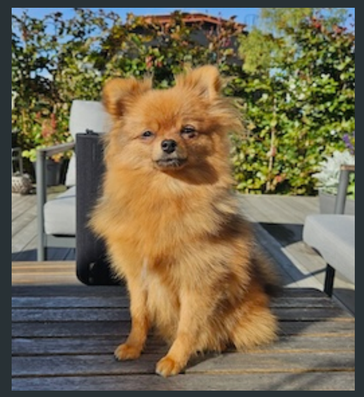
d)



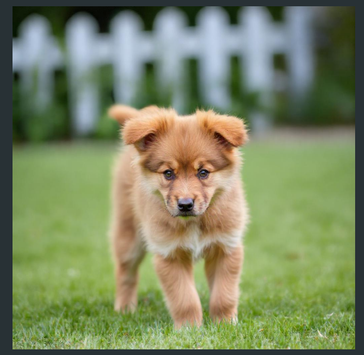
the AI generated images are prompted in model **flux. 1 schnell**

where is **Sammi?**

a)

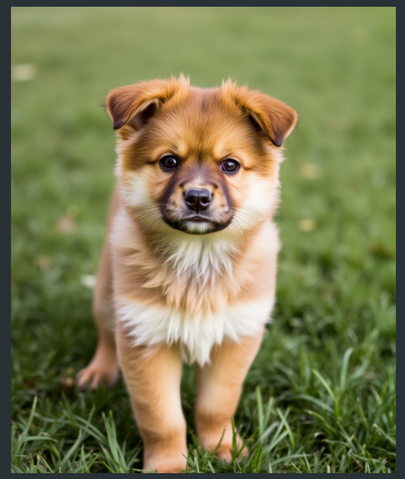


c)



b)

d)



the AI generated images are prompted in model **flux. 1 schnell**

what is AI and give a definition, we prompt ChatGPT 4.5

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by computer systems. These processes include:

- Learning (acquiring knowledge or skills through data, experience, or instruction)
- Reasoning (applying logic to reach conclusions or solve problems)
- Perception (interpreting sensory input, like vision or speech recognition)
- Natural language processing (understanding and generating human language)
- Decision-making (selecting actions based on learned or programmed criteria)

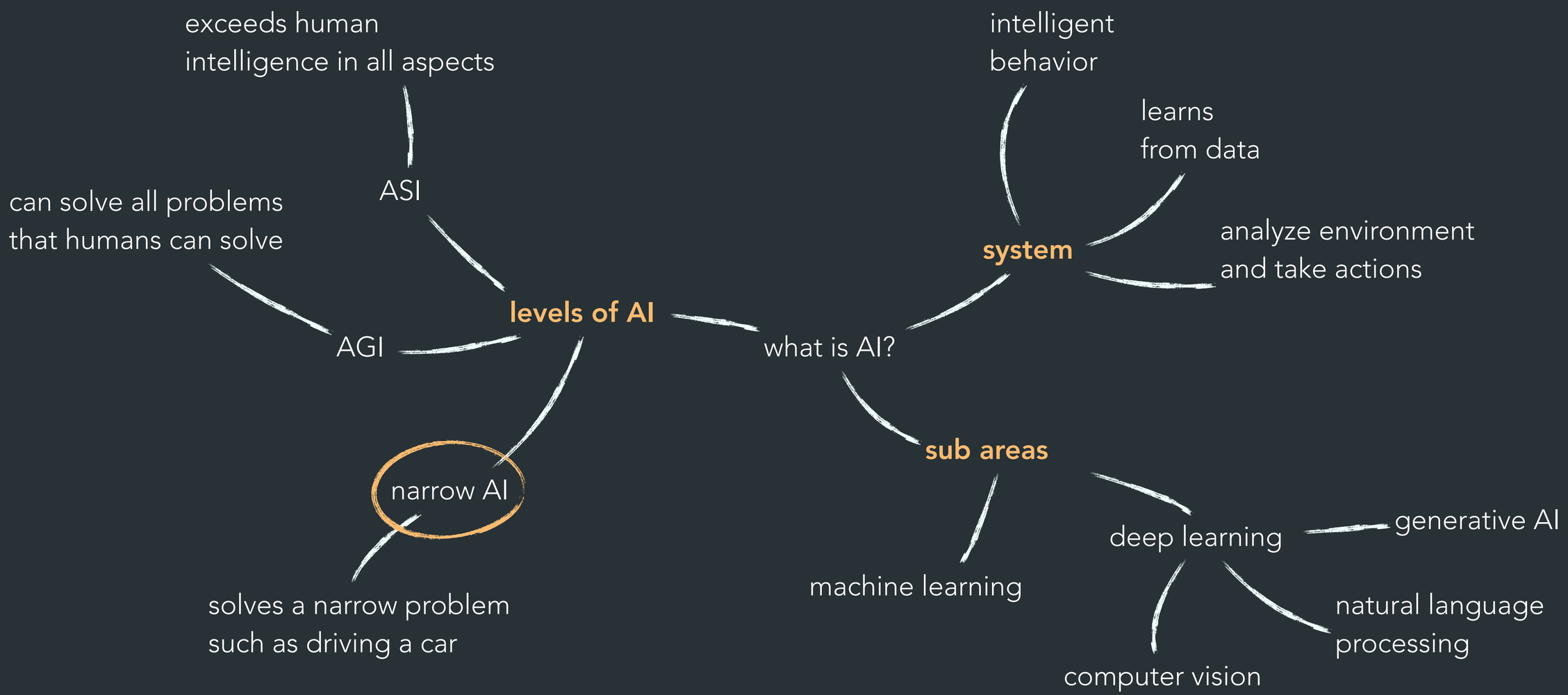
In essence, AI aims to create machines or software capable of performing tasks that typically require human intelligence.

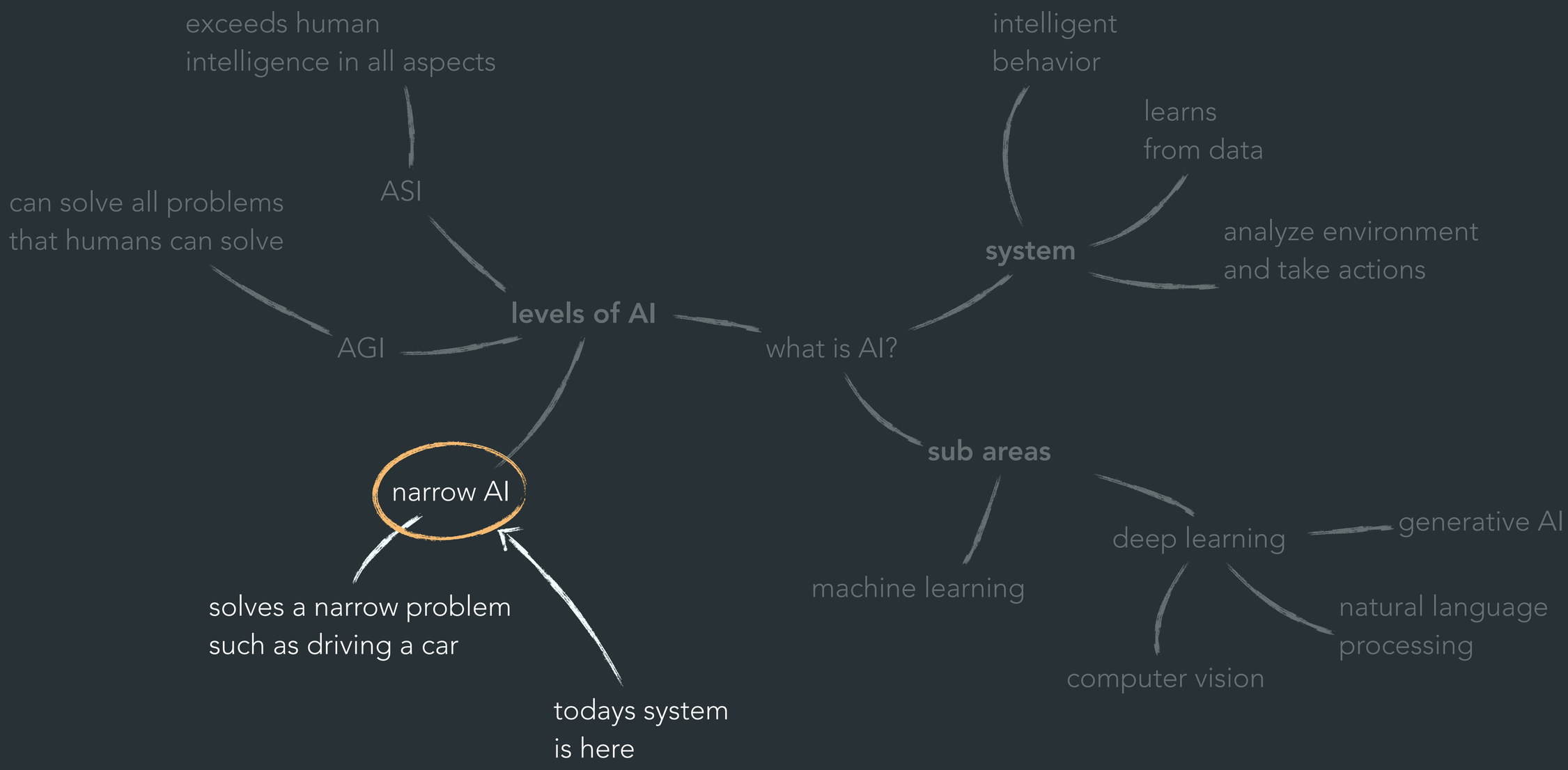
what is AI and give a definition, we prompt ChatGPT 4.5

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by computer systems. These processes include:

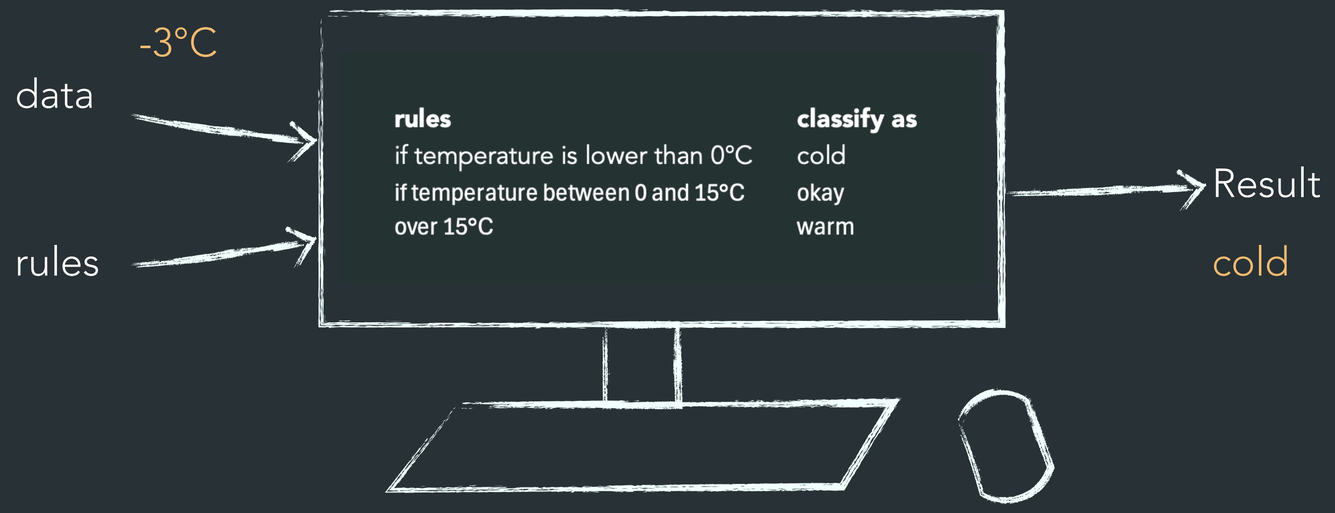
- Learning (acquiring knowledge or skills through data, experience, or instruction)
- Reasoning (applying logic to reach conclusions or solve problems)
- Perception (interpreting sensory input, like vision or speech recognition)
- Natural language processing (understanding and generating human language)
- Decision-making (selecting actions based on learned or programmed criteria)

In essence, AI aims to create machines or software capable of performing tasks that typically require human intelligence.


exceeds human
intelligence in all aspects
ASI
can solve all problems
that humans can solve
AGI
levels of AI
narrow AI
solves a narrow problem
such as driving a car
intelligent
behavior
learns
from data
system
analyze environment
and take actions
what is AI?
sub areas
machine learning
deep learning
generative AI
computer vision
natural language
processing


exceeds human
intelligence in all aspects
ASI
can solve all problems
that humans can solve
AGI
levels of AI
narrow AI
solves a narrow problem
such as driving a car
todays system
is here
intelligent
behavior
learns
from data
system
analyze environment
and take actions
what is AI?
sub areas
machine learning
deep learning
generative AI
computer vision
natural language
processing

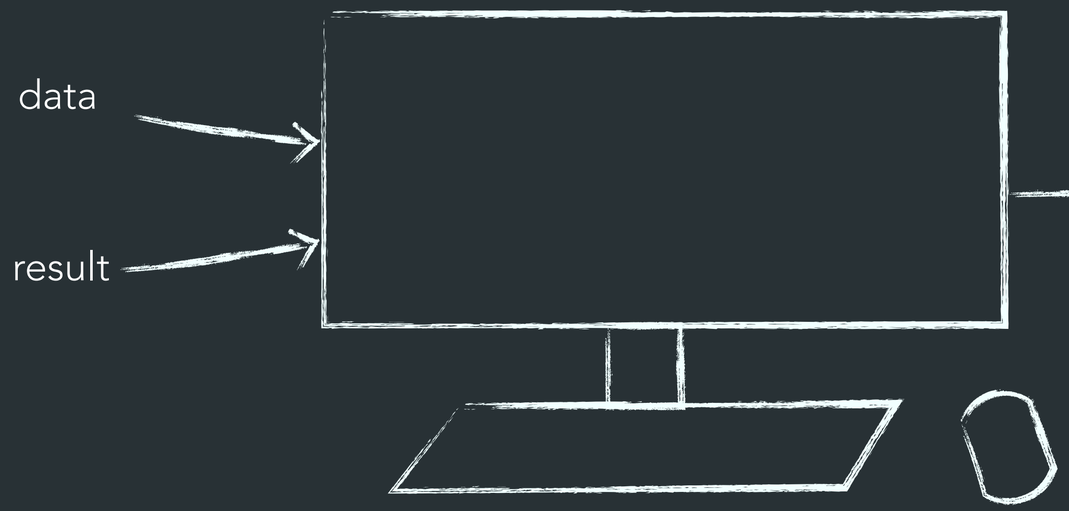
example: traditional **rule-based** system for temperature classification



developer (human) specifies the rules in beforehand

the rules become more and more complex the more factors we have, e.g. wind speed, air moisture, persons age, body fat etc

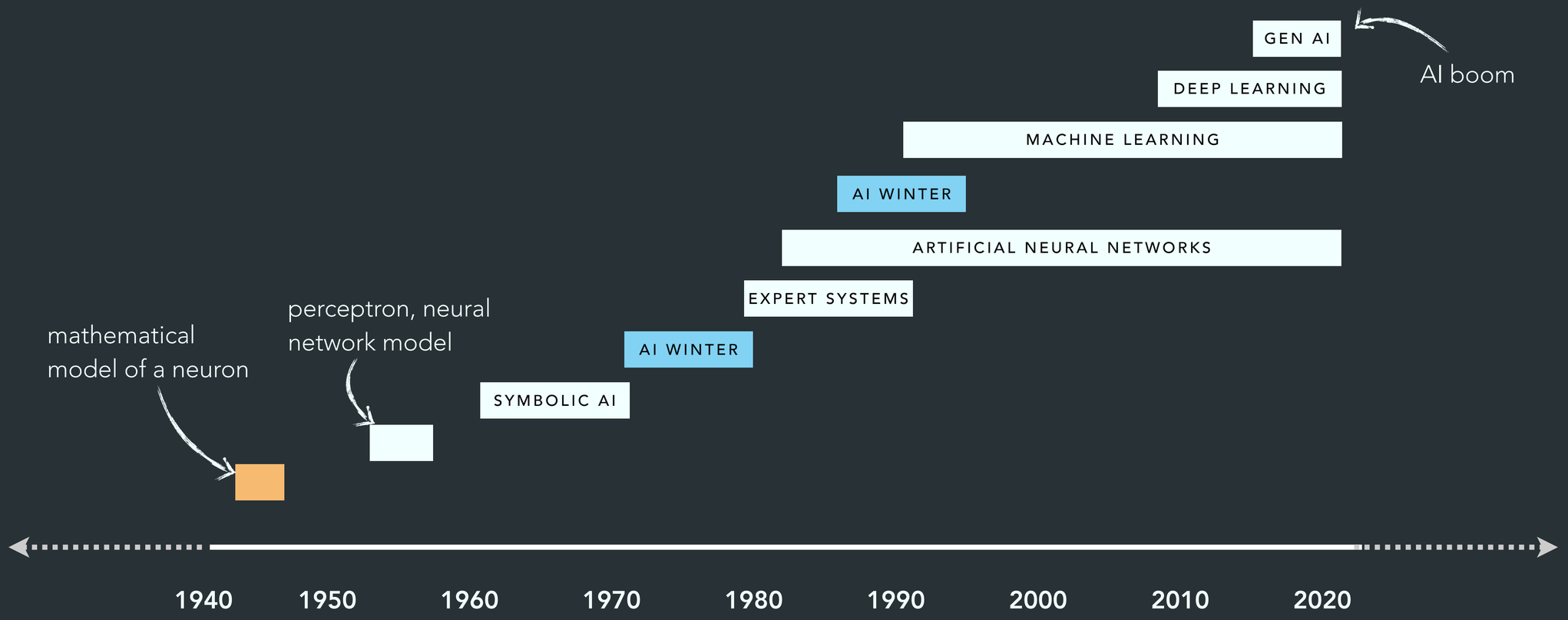
an **AI-based** system for temperature classification



rules	**classify as**
if temperature is lower than 0°C	cold
if temperature between 0 and 15°C	okay
over 15°C	warm

data and results is fed into an AI algorithm and it learns how to classify (rules)

a **historical timeline** over development of AI



it was through **machine learning** that
algorithms started to learn from data

before was rule based and expert systems
statistical methods
learns pattern in the data
flourished during 1990s but still widely used
today

genom **deep learning** kan algoritmerna
lära sig mer komplexa samband

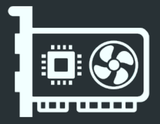
artificial neural networks
inspired by the brain, not a model of the brain
layers of neurons
many important algorithms during 80-90s
popular during 2010s

successes with **deep learning** during 2010s possible thanks to

algorithms from 80s required massive amounts of data and computing power

large amounts of data

computing power from GPUs



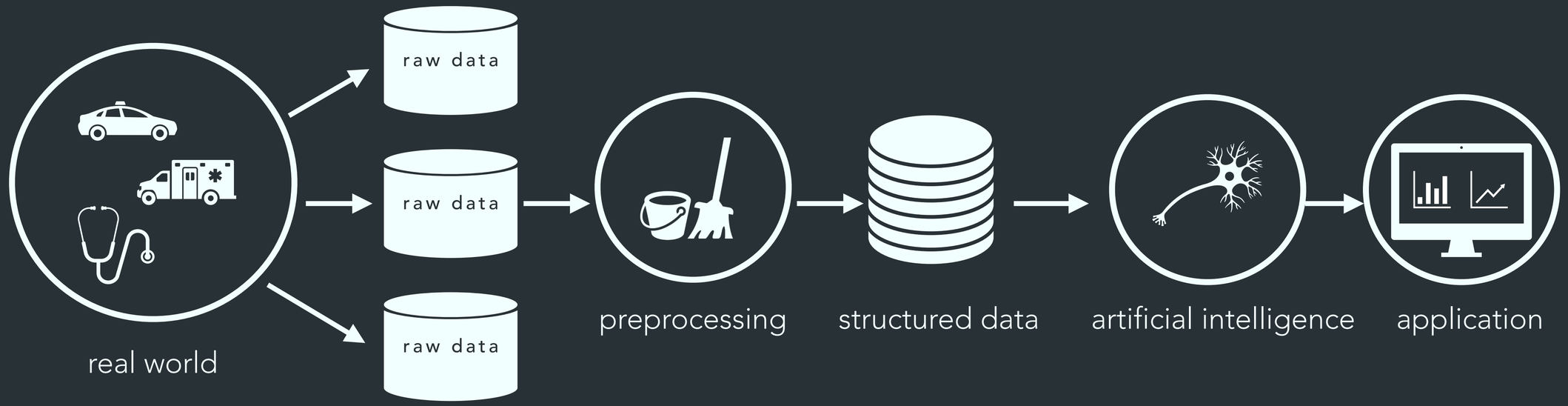
success storys in computer vision

with **generative AI** machines can generate content

AI-genererad bild med Dall-E 3

generate images, videos, sound, texts
large language models (LLM) based on deep learning, reinforcement learning and other techniques
AI-boom in 2020s

key to an AI-project is **data**



important to find **value** in an AI implementation

identify a problem that can give value when solved

why do we need AI here? Or are we implementing AI because our competitors are?

can the problem be solved without AI? If so don't force an AI solution

start with lowly hanging fruits as these can win trust within an organization

collaborate between domain experts and AI experts